



Effect of rotor aspect ratio and solidity on a straight-bladed vertical axis wind turbine in three-dimensional analysis by the panel method



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ABSTRACT

Due to the complicated flow field and aerodynamic forces characteristics, the performance and safety standard of straight-bladed VAWT have not been fully developed. The objective of this study is to investigate the effect of rotor aspect ratio and solidity on the power performance in three-dimensional analysis by panel method. The panel method is based on the assumption of an incompressible and potential flow coupled with a free vortex wake. First of all, the fluctuations of power coefficient and the circulation amount distribution of the bound vortex are discussed at the fixed solidity of $\sigma = 0.064$ during rotation. Then, the fluctuations of power coefficient and the circulation amount ratio are also investigated in the spanwise direction of the blade. It can be observed from the results that the peak of power coefficient increases with the increase of the ratio of the diameter and blade span length H/D at the fixed solidity. However, the optimum tip speed ratio was expected to be increased with the increase of H/D . Moreover, in the case of the fixed rotor aspect ratio of $H/c = 6$, the power coefficient depends on the rotor aspect ratio, rather than the ratio of the diameter and blade span length. Compared with the $H/D = 1.2$, the circulation amount ratio of $H/D = 0.9$ indicates a large negative value in the blade center position.

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1. Introduction

Wind energy has attracted worldwide interest because of global warming and environmental pollution resulted from fossil fuel consumption [1–5]. There has been a rapid growth of the applications of small straight-bladed VAWTs have been increasing rapidly in the urban environment due to their low sound emission and no-yaw control mechanism [3,6–9]. Many factors of the aerodynamic forces, including the varying wind velocities and directions and turbulence intensity, can impact the reliability of the wind turbine [7,10–13]. Therefore, these uncertainties of these factors should be considered in the development of design and optimization of wind turbine.

The relative wind velocity and angle of attack relative to the blade always change. They may induce the dynamic stall during the rotation. This phenomenon can cause flow separation, reattachment and vortex shedding from the blade surface, affecting large fluctuations of lift and drag coefficients for the airfoil [6,8,11,14–18]. In addition, the straight-bladed VAWTs with high aspect ratios are

subjected to large bending moments due to tangential forces, which may result in the airfoil failure [19]. Therefore, the flow field and aerodynamic forces characteristics of small straight-bladed VAWTs are rather complicated, especially the predictions of the vortex and power coefficients in the spanwise direction of the blade. Moreover, since the rotor aspect ratio and solidity variations of a straight-bladed VAWTs cause Reynolds number variations, any fluctuations of the power coefficient should be considered to investigate how they affect the wind turbine performance and flow field.

Up to now, several numerical and wind tunnel experimental studies related to the detailed flow field and aerodynamic prediction of VAWTs have been conducted. To investigate the flow field characteristics, Ferreira et al. [20] used the Detached Eddy Simulations (DES) model to predict the generation and shedding convection of vorticity. In their study, the visualization of the shedding and convection of the vorticity proved to be more specific for validation than aerodynamic force acting on the blade, especially in the location of the leading edge vorticity and the reattachment moment of the trailing edge vorticity. This has been supported by the studies of Tescione et al. [21], Hofemann et al. [22] as well as Dixon [23]. Then, Ferreira et al. [24] further studied the effect of

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Nomenclature

c	Airfoil chord length [m]
C_p	Power coefficient
C_{lp}	Local power coefficient
C_D	drag coefficient
C_L	Lift coefficient
$C_{L,invicid}$	Local lift coefficient
C_Γ	Circulation amount coefficient
D	Rotor diameter of wind turbine [m]
H	Span length of blade [m]
$l_{invicid}$	Local lift of per unit length [N]
N	Number of blade
N_{BLADE}	Number of each blade vortex panel
N_{WAKE}	Number of wake vortex panel
Q	Rotor torque [N·m]
U_0	Free stream wind velocity [m/s]

U_b	Resultant velocity to blade [m/s]
Δu	Induced velocity [m/s]
x	Longitudinal coordinate [m]
y	Lateral coordinate [m]
z	Vertical coordinate [m]
Γ	Circulation amount [m/s]
α_{eff}	Effective angle of attack [°]
α_g	Geometric angle of attack [°]
α_M	Corrected angle of attack [°]
α_{ind}	Induced angle of attack [°]
β	Blade pitch angle [°]
ε	Viscosity core radius
λ	Tip speed ratio
θ	Azimuth angle [°]
ρ	Air density [kg/m ³]
σ	Solidity

three dimensions on the development of vortices with Particle Image Velocimetry (PIV) in wind tunnel experiments. They found that the evolution inside the rotor volume of the wake was generated during the upwind blade passage where the tip vortex moves inboard. Maeda et al. [25] and Li et al. [26] investigated the effects of tip speed ratio on the flow field around a two-bladed straight-bladed VAWT with an airfoil of NACA0021 using a Laser Doppler Velocimeter (LDV) technology in wind tunnel experiments. The authors found that the area of low wind velocity expanded with the increase of tip speed ratio, and the reverse flow was generated at some downstream regions with high tip speed ratio. Li et al. [27] further investigated the fluctuations of circulation amount using $k-\varepsilon$ Shear Stress Transport turbulence model and induced velocities through LDV measurement in the spanwise direction. From this study, it was found that the circulation amount showed the largest value at the blade center height and the smallest value at the blade tip. Similarly, the induced velocity illustrated the maximum value at the blade center height and decreased with the increase of spanwise position. After that, Li et al. [28] also discussed the wake characteristics by three-cup type anemometers in the natural environment. Compared with the results from wind tunnel experiments, they found that the wake velocities in field experiments had a quicker recovery than those from wind tunnel experiments. Rolin et al. [29] measured the wake of VAWT with PIV in a turbulent boundary layer at low Reynolds numbers and low tip speed ratios in wind tunnel experiments. From their research, it was indicated that the boundary layer effects might be responsible for the lower velocity deficit in the core of the wake. Danao et al. [30] studied on the effects of steady and unsteady wind on the performance of VAWT using RANS-based CFD. From their research, the importance of stall and flow re-attachment on the performance of the turbine with unsteady winds was discussed in detail.

To have a better understanding on the aerodynamic force characteristics in the three-dimensional unsteady flow, Qin et al. [31] developed a CFD model for the evaluation of aerodynamic forces of a straight-bladed VAWT with three-blades, based on the unsteady Reynolds averaged Navier-Stokes equations. In this study, the maximum torque curve in the three-dimensional was similar to the two-dimensional one, but was moved downwards, resulting in a significantly lower average torque produced by about 40%. Castelli et al. [32], El-Samanoudy et al. [33], Howell et al. [34] and Li et al. [9] investigated the effect of number of blades on the power coefficient of straight-bladed VAWTs by wind tunnel experiments or CFD

simulations. They found that the maximum power coefficient decreased with the increase of number of blades, while the tip speed ratio was shifted to a lower value. El-Samanoudy et al. [33] also studied the effect of pitch angle on the maximum power coefficient with CFD simulations. They found that the pitch angle did not have a significant impact on the power coefficient of wind turbine when the numbers of blades were the same. Their results were also supported by parametric experiments or simulations by Li et al. [9,25], Fiedler and Tullis [35], and Hwang et al. [36]. To determine the effect of wind velocity on the power performance of straight-bladed VAWTs, Howell et al. [34] focused on the VAWT performance of a small scale three-bladed VAWT as a function of the tip speed ratio with an experimental and computational analysis. For the wind velocities of $U = 3.16$ m/s, 4.31 m/s, 5.07 m/s and 5.45 m/s, the maximum values of power coefficient were about $C_p = 0.112, 0.173, 0.181$ and 0.223 , respectively. The literatures of Danao et al. [37] Eboibi et al. [38] and Sunyoto et al. [39] also indicated that the wind turbine with a higher wind velocity had a higher optimum power coefficient. Li et al. [40] measured the power performance of airfoil NACA 0021 for the Reynolds numbers between 1.85×10^5 and 2.89×10^5 in wind tunnel experiments. From this study, it was noted that the maximum power coefficient increased with the increase of the Reynolds number. The observations of this result matched well with the studies of Tirkey et al. [41], Roh et al. [42] and Tai et al. [43]. Meanwhile, the effect of solidity on VAWT depending on the different numbers of blade was compared in this wind tunnel experiment. The maximum power coefficient decreased with the increase of solidity. However, the maximum torque coefficient increased with the increase of solidity. Furthermore, Li et al. [29] further examined the pressure acting on the rotor surfaces of VAWT in the spanwise direction of $z/(H/2) = 0, 0.55, 0.70$ and 0.80 , respectively. They pointed out that the fluid force decreased with the increase of spanwise positions excluding the position of support structure. Chen et al. [44] constructed an analytical model on the basis of $K-\varepsilon$ turbulence model to study the effect of blade chord lengths on the performance of a straight-bladed VAWT. It was noted that the increase of the chord length played great role in improving the power performance. The similar conclusion was also drawn by Mohamed [45] and El-Samanoudy et al. [33].

The reviews and discussion above strongly suggest that great achievements have been acquired for the understanding of flow field and aerodynamic forces characteristics of straight-bladed VAWTs. However, very limited literature is available for the

investigations on the effects of the blade spanwise positions on the performance during rotation. Therefore, this paper aim to discuss the power and vortex characteristics with panel method to evaluate the effects of the rotor aspect ratio and solidity (different chord lengths and blade spans) on the performance of straight-bladed VAWTs.

2. Panel method

In the present work, special attention is given to the issue of development of the vortex characteristics, following the observations and conclusions of the work of Li et al. [46] in 3D. The panel method is based on the assumption of an incompressible and potential flow coupled with a free vortex wake. Previous research (Voutsinas and Riziotis [47], Zanon et al. [48] and Tescione et al. [49]) showed that this approach was to predict the aerodynamic load prediction in good agreement with the experimental results. Meanwhile, the panel method is more convenient than CFD in the analysis of three-dimensional unsteady flow field, because it does not require any computational grids.

2.1. Induced velocity of vortex

In this study, the induced velocity of vortex is calculated by the Rankine vortex model. In the Rankine vortex model, the vicinity of the vortex coordinate system is as a forced vortex and the far away area from the vortex coordinate system is as a free vortex. Fig. 1 indicates the distribution of induced velocity which is induced from one vortex coordinate system of the circulation amount Γ . The vertical distance from the vortex coordinate system to the boundary region between the free vortex area and the forced vortex area is called viscous core radius ε .

2.2. Fluid force acting on the blade

Fluid force acting on the blade is obtained from the theorem of Kutta-Joukowski [50]. Local lift coefficient $C_{L,invicid}(z_b)$ generated from the chordwise blade vortex panel in the blade span direction position of z is determined by the following equation:

$$C_{L,invicid}(z_b) = \frac{l_{invicid}(z_b)}{\frac{1}{2}\rho c(z_b)U_b^2} \quad (1)$$

where ρ is the air density, c is the airfoil chord length, U_b is the

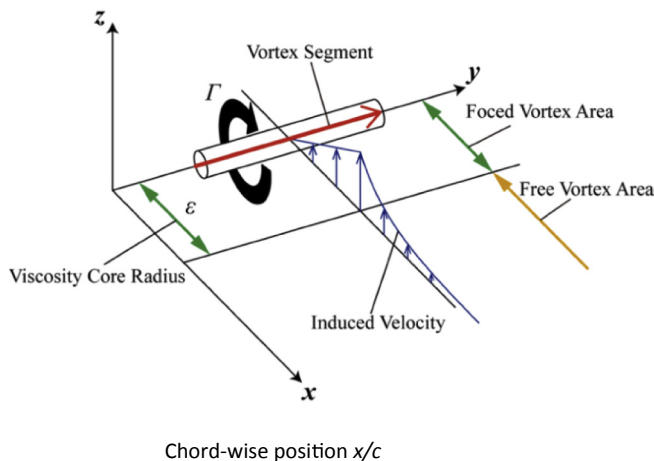


Fig. 1. Distribution of induced velocity which is induced from one vortex coordinate system of the circulation amount Γ .

resultant flow velocity to the rotor blade in the blade coordinate system of $x_b y_b$ plane and $l_{invicid}$ is the local lift of per unit length at the inviscid calculation.

In general, the angle of attack is defined as an angle between the uniform inflow and the chord line in the two-dimensional flow. Hence, it is difficult to determine the angle of attack of wind turbine in the three-dimensional flow. Fig. 2 shows the relationship of the resultant inflow velocity and the inflow angle to the blade. Where, β is the blade pitch angle, α_g is the geometric angle of attack which is defined by the resultant inflow and the blade pitch angle, α_{ind} is the decrease of the angle of attack by the induced velocity (induced angle of attack), and α_{eff} is the effective angle of attack which is represented by the following equation [51]:

$$\alpha_{eff} = \alpha_g - \alpha_{ind} \quad (2)$$

2.3. Viscosity correction by the blade characteristics data

In the present analysis, the angle of attack in the blade spanwise section is calculated by the $C_{L,invicid}(z_b)$ without considering viscosity in each of blade spanwise position of the physical panel. As shown in Fig. 3 ①, the angle of attack $\alpha(z_b)$ can be obtained by $C_{L,invicid}(z_b)$ from the primary function [18].

$$\alpha(z_b) = \frac{C_{L,invicid}(z_b) - C_{L\alpha=0}}{\Delta C_L} = \frac{1}{\Delta C_L} \left\{ \frac{l_{invicid}(z_b)}{\frac{1}{2}\rho((u_{b,0} + \Omega y_b)^2 + w_{b,0}^2)c(z_b)} - C_{L\alpha=0} \right\} \quad (3)$$

where, $C_{L\alpha=0}$ is the lift coefficient at the zero angle of attack, $u_{b,0}$ is the uniform flow velocity, and $w_{b,0}$ is the induced velocity generated from the trailing vortex panel. After the non-viscous calculation process, it is essential to carry out the viscosity correction using a two-dimensional airfoil characteristics data. According to the angle of attack determined by Equ. (3), the lift coefficient $C_L(z_b)$ (Fig. 3 ②) and the drag coefficient $C_D(z_b)$ (Fig. 3 ③) can be calculated using the two-dimensional airfoil characteristic data in consideration of the influence of the viscous.

2.4. Simulation conditions

In this study, a straight-bladed VAWT in the wind tunnel experiment is analyzed. There are two blades and the airfoil type is

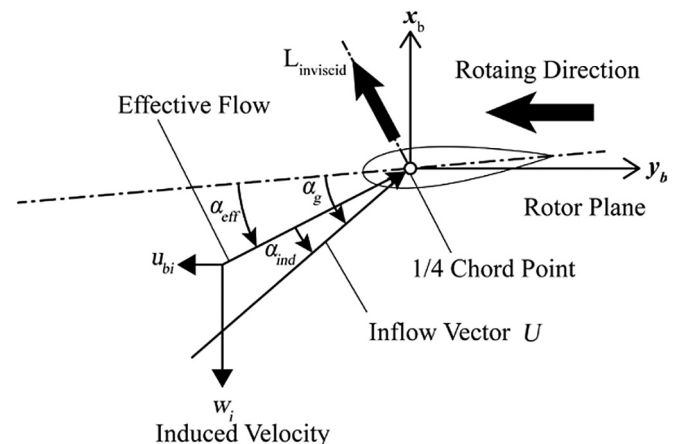


Fig. 2. Angle of attack relative to the rotor blade.

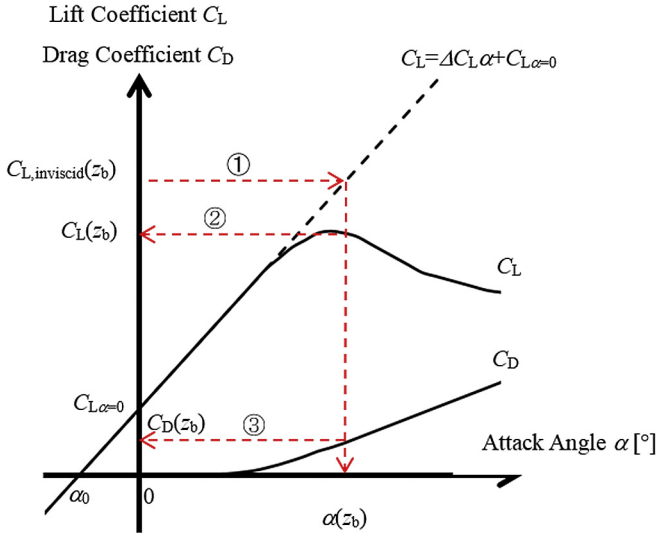


Fig. 3. Process of viscosity correction in this study.

NACA0021. Regarding the description of this wind turbine, more details of the functions and parameters can be found in literatures of Maeda et al. [52] and Li et al. [7,9,15,23,26,28,40]. Fig. 4 represents the main structures of VAWT, which are composed of blade, support structure and rotor shaft. Solidity σ , showing the a relation between the blade area and wind turbine swept area, and is defined as [35,40,42,53].

$$\sigma = \frac{Nc}{\pi D} \quad (4)$$

where N is the number of blades.

As described in Section 2.1, the rotor blade is divided by physical panels. A single blade is divided into 16 pieces in the chord

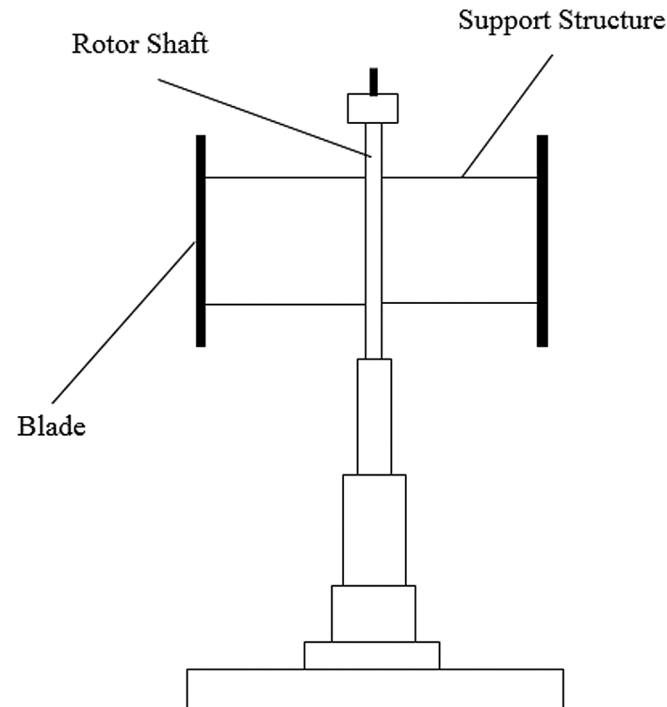


Fig. 4. Main structure of VAWT.

direction and 10 sheets in the spanwise direction. Therefore, the total number of physical panels is 160 at the whole analysis area. Table 1 summarizes the numerical conditions in the panel method. Analysis of the vortex field is carried out at the tip speed ratio obtained from the performance analysis. The circulation amount coefficient C_Γ at each blade section can be calculated by Refs. [29,51,54].

$$C_\Gamma = \frac{\Gamma}{cU_0} \quad (5)$$

In this study, collocation point is employed to determine the circulation amount Γ with the rotor blade. The induced velocity is calculated with all the vortex line elements at the collocation point. The circulation amount Γ of blade vortex can be shown as follows [47,54].

$$\sum_{m=1}^{N_{\text{BLADE}}} [B_m \Gamma_m + (\mathbf{W}_0 - \boldsymbol{\omega} \times \mathbf{r}_b)_m \cdot \mathbf{n}] + \sum_{n=1}^{N_{\text{WAKE}}} C_n \Gamma_n = 0 \quad (6)$$

where B_m and C_n are the influence coefficients determined by the position of blade vortex panel and wake vortex panel. N_{BLADE} and N_{WAKE} are the number of each blade vortex panel and wake vortex panel, respectively.

3. Results and discussion

There are multiple parameters affecting the performance of the straight-bladed VAWT, such as the airfoil shape, the rotor diameter and spanwise length et al. And the study of optimum design of this type wind turbine is still poor in addition to its complex fluid phenomena. The three-dimensional effect of the changing ratio of the diameter and blade span length on the fluid force generated in the blades of VAWT will be discussed in this section.

3.1. Fixed solidity

Fig. 5 shows how the fluctuations of power coefficient changes against the tip speed ratio λ for the fixed solidity of $\sigma = 0.064$, the changing ratio of the diameter and blade span length (H/D). The horizontal axis is the tip speed ratio λ and the vertical axis represents the power coefficient C_p , respectively. As shown in this figure, the peak of power coefficient increases with the increase of the ratio of the diameter and blade span length (H/D). When the ratio of the diameter and blade span length are $H/D = 0.4, 0.6, 0.9$ and 1.2 , the maximum power coefficients are about $C_p = 0.183, 0.223, 0.247$ and 0.273 , respectively. However, the optimum tip speed ratio is expected to increase with the increase of H/D . It means the maximum power coefficient is higher at a larger value of H/D . There is a large value of rotor aspect ratio when the H/D is larger. Effect of induced drag due to blade tip vortex is smaller with the increase of

Table 1
Numerical conditions with panel method in this study.

Parameters classification	Rotor diameter D [m]	Chord length c [m]	Blade span H [m]	Solidity σ	Rotor span diameter ratio H/D
Fixed solidity	3.0	0.3	1.2	0.064	0.4
	2.0	0.2	1.2		0.6
	2.0	0.2	1.8		0.9
	2.0	0.2	2.4		1.2
Fixed aspect ratio	2.0	0.2	1.2	0.064	0.6
	2.0	0.3	1.8	0.095	0.9
	2.0	0.4	2.4	0.127	1.2

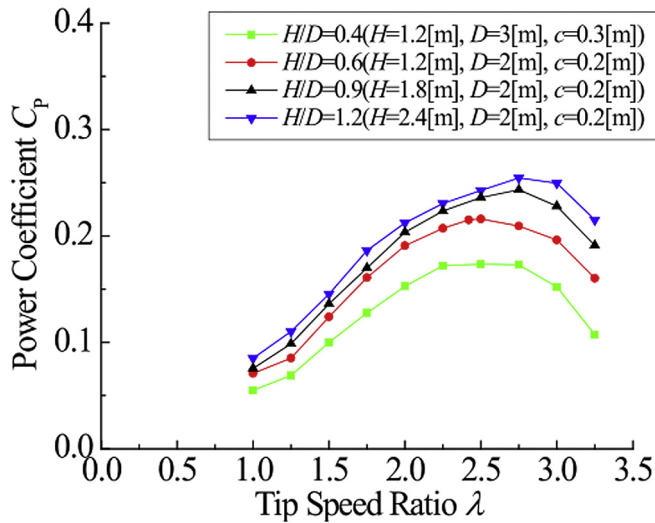


Fig. 5. Fluctuations of power coefficient are compared with a fixed solidity in different rotor aspect ratios, depending on the tip speed ratio λ . The ratio of the diameter and blade span length are $H/D = 0.4, 0.6, 0.9$ and 1.2 .

aspect ratio, so that the power performance of the rotor blade is improved.

Fig. 6 shows the circulation amount distribution of the bound vortex at the optimum tip speed ratio. In this figure, horizontal axis is the azimuth angle θ and the vertical axis is the blade spanwise cross-section, respectively. The color map indicates the non-dimensional circulation amount of bound vortex with the rotor blade. Fig. 6 (a), (b) and (c) describes the results of the ratios of the diameter and blade span length of $H/D = 0.4, 0.9$ and 1.2 , respectively. As noted in these figures, the circulation amount takes a maximum value at the upstream region of $\theta = 100^\circ$ and indicates a negative value at the downstream region of $180^\circ \leq \theta \leq 360^\circ$. The reason is that, the circulation amount gradient becomes larger and the blade tip vortex also becomes stronger at the upstream region, and circulation amount gradient becomes smaller and the blade tip vortex also becomes weaker at the downstream region. Moreover, it is found that the circulation amount decreases with the increase of blade spanwise position in any cases of the azimuth angle. In addition, the range of strong circulation amount becomes narrower and longer in the vertical direction with the increase of the H/D . The reason is that, there are large fluctuations of the tip vortex velocity, which is resulted from the blade tip. The rotor blade with a large aspect ratio is less susceptible to the influence of the resultant inflow velocity.

The fluctuations of the circulation amount ratio in the spanwise direction of the blade are compared in Fig. 7. In this figure, the circulation amount ratio shows the results of the azimuth angle of 100° in the bound vortex at the optimum tip speed ratio. The circulation amount ratio is the ratio of the circulation amount in the blade center section and that generated from the bound vortex in each blade spanwise cross-section. Horizontal axis indicates the circulation amount ratio, while the vertical axis indicates the cross-section in spanwise position. This figure illustrates that the circulation amounts show large values at the rotor center height, and are reduced with the increase of spanwise position. Moreover, the circulation amount ratio at a small value of H/D decreases in a wide range of spanwise position near the blade tip. Since the tip vortex is a free vortex, the induced velocity generated by the blade tip vortex is inversely proportional to the distance from the blade tip. Therefore, induced velocity of the blade tip vortex is smaller when closer to the blade tip. In the case of small H/D , the influence range

of the blade tip vortex is relatively larger than that of the high H/D .

To consider the three-dimensional effect on the power coefficient of VAWT, Fig. 8 represents the evolution of the local power coefficient C_{lp} in each blade cross-sectional position. In this figure, horizontal axis is the blade cross-sectional position $H/(Z/2)$ and the vertical axis is the local power coefficient C_{lp} , respectively. The green, red, black and blue values are obtained from the ratio of the diameter and blade span length $H/D = 0.4, 0.6, 0.9$ and 1.2 , respectively. It is confirmed that the local power coefficient reduces when closer to the blade tip for any cases of the aspect ratio. The power coefficient at the blade central position increases with the increase of the rotor aspect ratio. This is because the influence of induced velocity generated by the blade tip vortex in a small rotor aspect ratio is smaller than that of large rotor aspect ratio. And then, the wake caused by the tip vortex extends to a small value of angle of attack.

3.2. Fixed rotor aspect ratio

Fig. 9 illustrates the fluctuation of power coefficient curve against the tip speed ratio λ in the case of the fixed aspect ratio. In this figure, the rotor aspect ratio is $H/c = 6$ and the ratios of the diameter and blade span length are $H/D = 0.6, 0.9, 1.2$, respectively. From the figure, there is no significant fluctuation of the maximum value of power coefficient in any cases of H/D . It should be noted that the optimum tip speed ratio is smaller at a higher H/D because of a higher solidity. Therefore, the power coefficient depend on the rotor aspect ratio, rather than the ratio of the diameter and blade span length. Furthermore, the rotor aspect ratio determines the optimum tip speed ratio.

Fig. 10 (a) and (b) describe the fluctuations of the circulation amount ratio of the bound vortex in the spanwise direction of the blade at the optimum tip speed ratio. Fig. 10 (a) and (b) show the results of $H/D = 0.9$ and 1.2 , respectively. As shown in the two figures, the fluctuations of the circulation amount ratio of the bound vortex in the blade spanwise direction are similar to H/D on the upstream of azimuth angle position. However, the fluctuations have changed significantly on the downstream of the azimuth angle position. Compared with the $H/D = 1.2$, the circulation amount ratio of the bound vortex at the case of $H/D = 0.9$ indicates a strong negative circulation amount in the blade center position. The thrust of $H/D = 0.9$ which has a larger solidity is smaller than that of the $H/D = 1.2$. Therefore, the deceleration amount of flow is reduced when it passes through the upstream side of the rotor blade. The resultant inflow to rotor blades passing through the downstream side becomes larger, so the negative circulation amount becomes stronger.

Fig. 11 represents the circulation amount ratio of the bound vortex in the spanwise direction of the blade under the same conditions of Fig. 9. It shows the results of azimuth angle that indicates the circulation amount peak at the optimum tip speed ratio for each case. From this figure, it can be observed that the circulation amount ratios have similar values for any cases of the H/D in the spanwise direction of the blade. In addition, the circulation amount ratio is slightly changed by H/D in the position closest to the blade tip. In the case of the same aspect ratio, the rotor aspect ratio is only affected by the flow in the vicinity of blade tip. Moreover, the impact of the three-dimensional flow in Fig. 11 is smaller than the blade tip vortex in Fig. 9. Therefore, three-dimensional effect of the rotor blade is largely affected by the fluctuation of rotor aspect ratio when the ratio of the diameter and blade span length is changed.

Fig. 12 compares the fluctuations of local power coefficient against the tip speed ratio λ at the blade cross-sectional position under the same conditions of Fig. 9. As shown in this figure, the

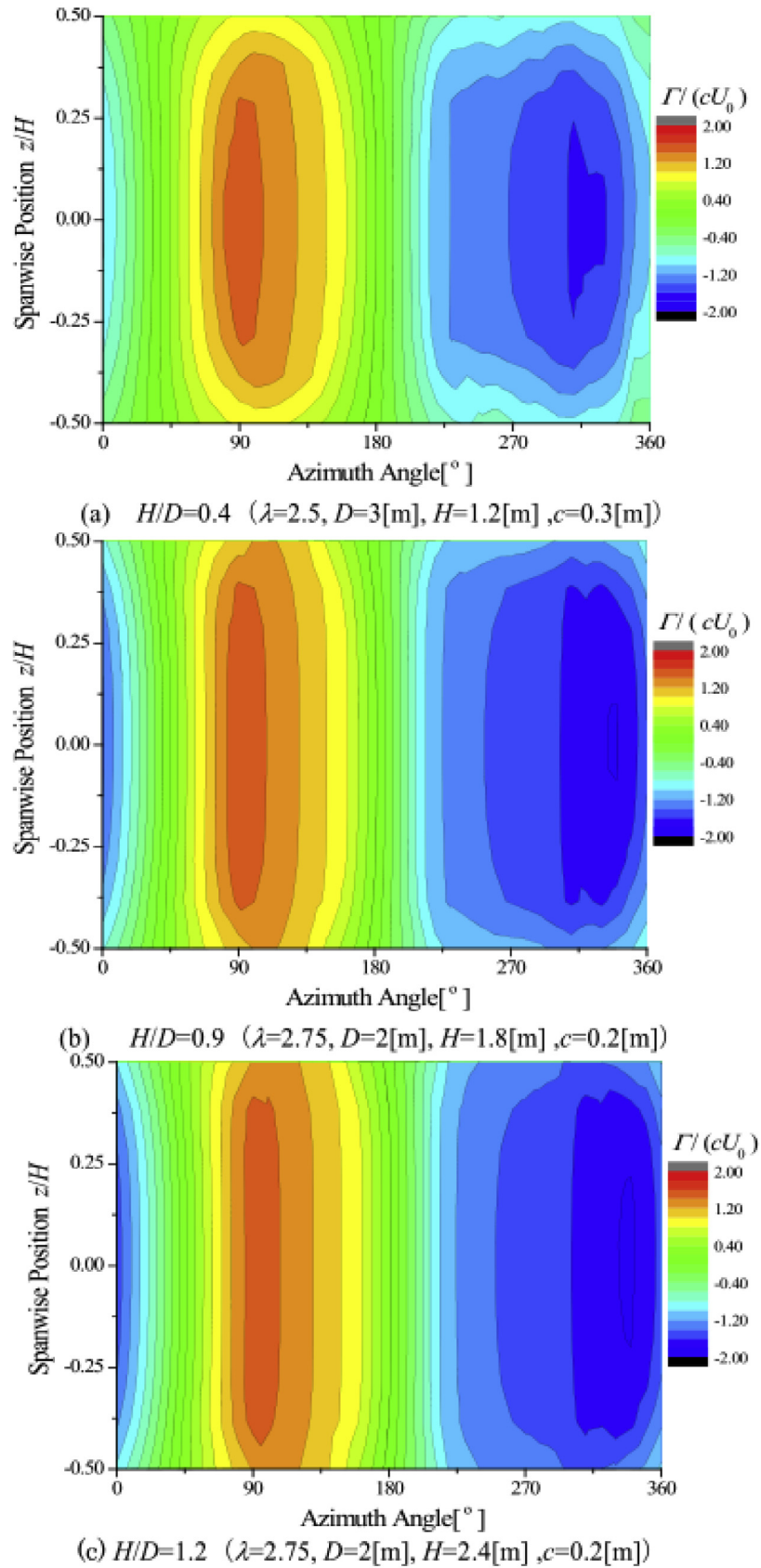


Fig. 6. Circulation distribution of the bound vortex with a fixed solidity in different rotor aspect ratios, depending on azimuth angle at the optimum tip speed ratio. Figure (a), (b) and (c) describe the results of the ratios of the diameter and blade span length of $H/D = 0.4, 0.9$ and 1.2 , respectively.

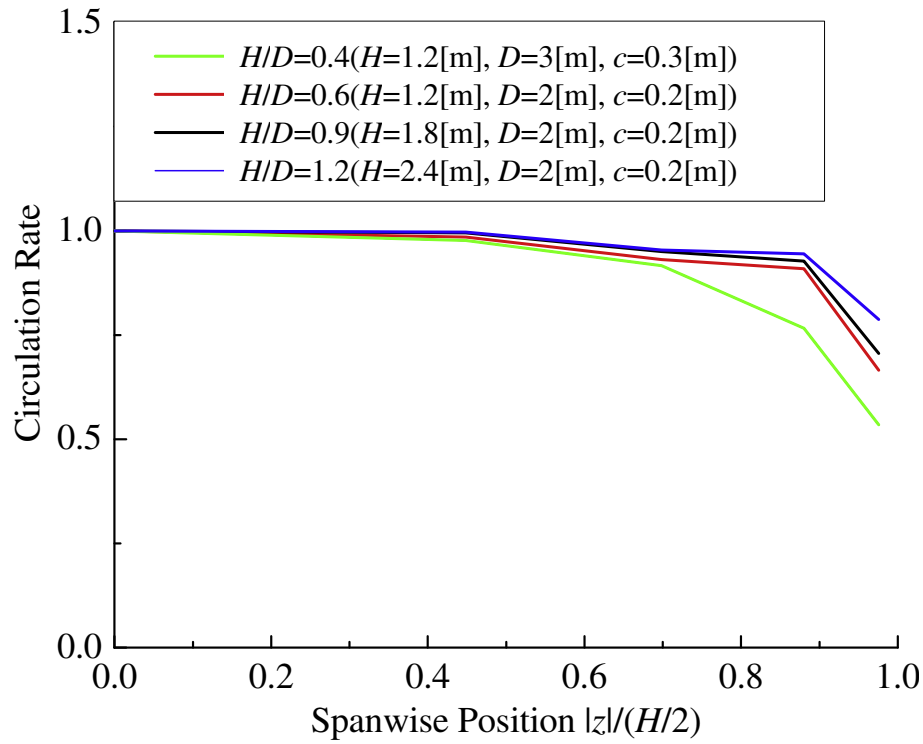


Fig. 7. Fluctuations of circulation rate with a fixed solidity in different rotor aspect ratios, depending on the spanwise position.

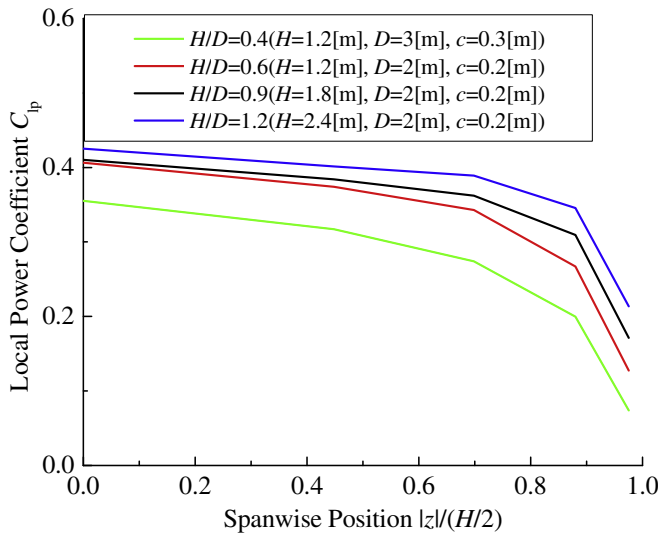


Fig. 8. Fluctuations of local power coefficient are compared with a fixed solidity in different rotor aspect ratios, depending on the spanwise position.

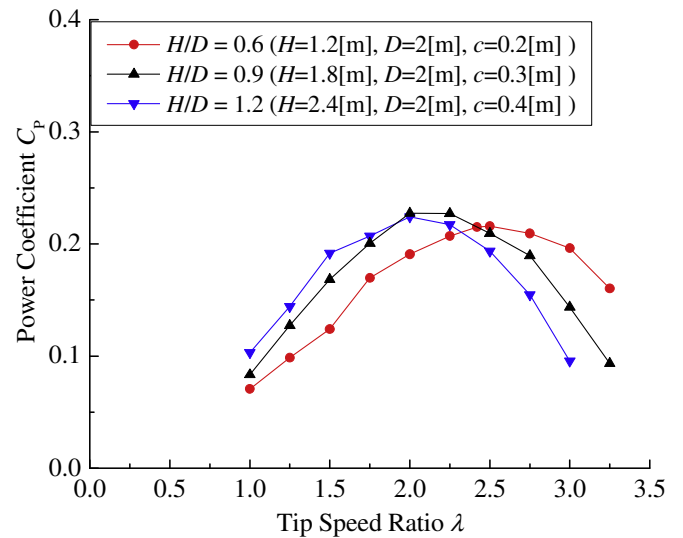


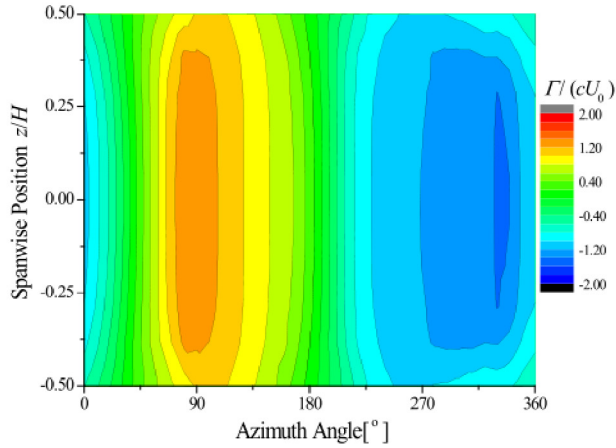
Fig. 9. Fluctuation of power coefficient curve against the tip speed ratio λ in the case of the fixed aspect ratio.

results of local power coefficient of $H/D = 0.9$ indicate a good agreement with the values of $H/D = 1.2$. This is because in the cases of the $H/D = 0.9$ and 1.2 , the circulation amount ratio have a similar tendency at the upstream side of azimuth angle $\theta = 0-180^\circ$ which generates a large value of tangential force. Furthermore, in each condition, the local power coefficient is approximately equal in the vicinity of blade tip for any cases of H/D . However, the local power coefficient shows a large value at the case of $H/D = 0.6$. From Fig. 11, it can be seen that the circulation amount ratio of the bound vortex is greater in the vicinity of the azimuth angle of $\theta = 100^\circ$, when the ratio of the diameter and blade span length is $H/D = 0.6$. Therefore,

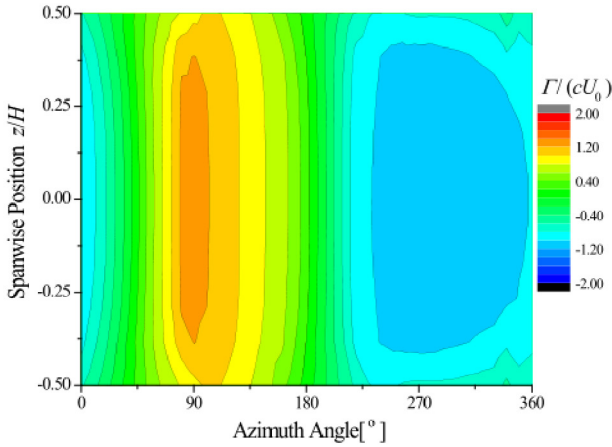
the large difference of the power coefficients is believed to be caused in the vicinity of blade center. This result indicates that the power coefficient of the solidity has a more significant effect than that of the rotor aspect ratio.

4. Conclusions and future research

In this study, the effects of the blade spanwise positions on the performance during rotation were investigated by panel method. The objective is to discuss the power and vortex characteristic to evaluate the effects of the rotor aspect ratio and solidity on the performance of



(a) $H/D=0.9$ ($\lambda=2.0$, $D=2$ [m], $H=1.8$ [m], $c=0.3$ [m])



(b) $H/D=1.2$ ($\lambda=2.0$, $D=2$ [m], $H=2.4$ [m], $c=0.4$ [m])

Fig. 10. Fluctuations of the circulation amount ratio of the bound vortex in the spanwise direction of the blade at the optimum tip speed ratio. (a) and (b) show the results of $H/D = 0.9$ and 1.2 , respectively.

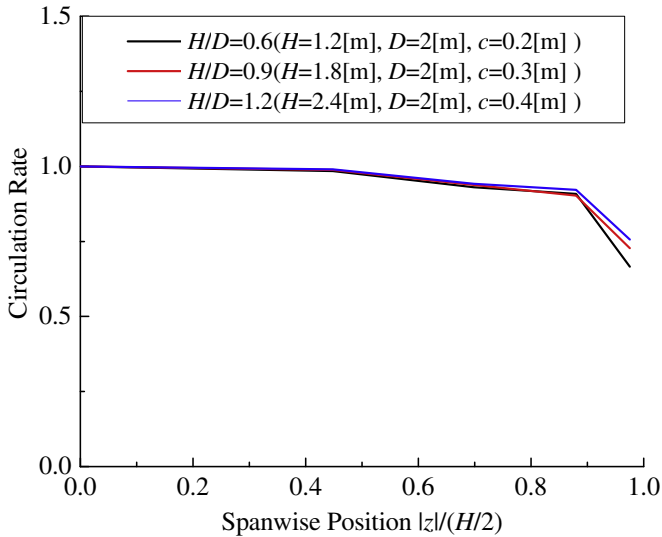


Fig. 11. Fluctuations of the circulation amount ratio of the bound vortex in the spanwise direction of the blade. The ratio of the diameter and blade span length are $H/D = 0.6$, 0.9 and 1.2 , respectively.

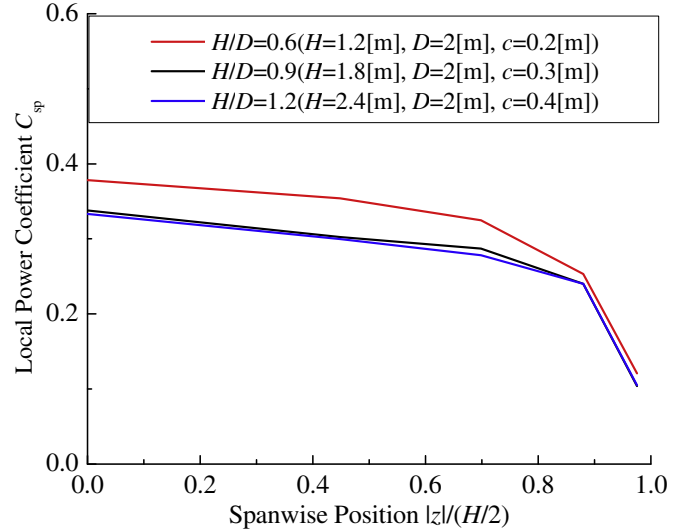


Fig. 12. Fluctuations of the local power coefficient at each spanwise position. The ratio of the diameter and blade span length are $H/D = 0.6$, 0.9 and 1.2 , respectively.

straight-bladed VAWTs in the different ratios of the diameter and blade span length $H/D = 0.40$, 0.60 , 0.90 and 1.20 , respectively. Through comparisons of the power coefficient and vortex characteristic, the following information were discussed in details.

For the fixed solidity of $\sigma = 0.064$, the peak of power coefficient increased with the increase of the ratio of the diameter and blade span length (H/D). However, the optimum tip speed ratio was expected to be increased with the increase of H/D . Moreover, the circulation amount took a maximum value at the upstream and indicated a negative value at the downstream. The circulation amount decreased with the increase of blade spanwise position. The circulation amount ratio at a small value of H/D represented a decrease in a wide range of spanwise position near the blade tip. Furthermore, the power coefficient at the blade central position increased with the increase of the rotor aspect ratio.

For the fixed rotor aspect ratio of $H/c = 6$, the power coefficient was dependent on the rotor aspect ratio, rather than the ratio of the diameter and blade span length. Compared with the $H/D = 1.2$, the circulation amount ratio of the bound vortex at the case of $H/D = 0.9$ indicated a strong negative circulation amount in the blade center position. The circulation amount ratios had similar values for any cases of the H/D in the spanwise direction of the blade. In addition, in the case of the same aspect ratio, the rotor aspect ratio was only affected by the flow in the vicinity of blade tip. The power coefficient of the solidity had a more significant effect than that of the rotor aspect ratio.

The numerical investigation presented in this paper provides a better understanding of the effect of rotor aspect ratio and solidity on the power coefficient and vortex characteristic of VAWT. However, the flaw is that all the summaries are clarified by numerical simulations. Therefore, in our future research, the wind tunnel measurements should be conducted to make the job more completed. Meanwhile, the dynamic stall will also be studied by PIV experiments and simulations.

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